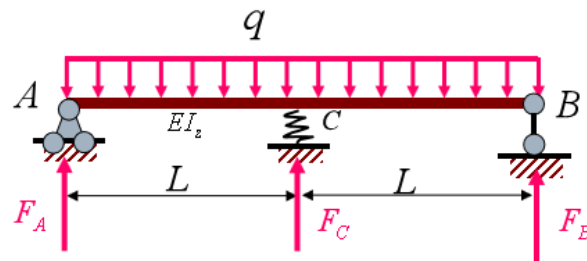


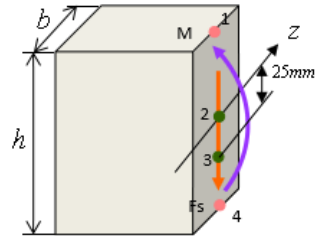
Submission: Please submit the scanned/typed document to Canvas.

1. As shown, a simply supported beam AB with Young's modulus E and moment of inertia I_z carries a uniform distributed load q . An elastic (spring) support with stiffness k is added at the midpoint C . If the bending moment at section C is zero, determine the spring stiffness k .



2. For a rectangular cross-section beam, the shear force on the section is $F_s = 120$ kN, and the bending moment is $M = 10$ kN · m. Draw the stress element at points 1, 2, 3, and 4 on the cross-section to represent their stress states and determine the principal stresses at each point.

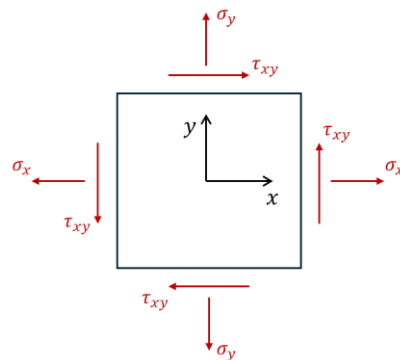
point. $b = 60$ mm, $h = 100$ mm. (Hint: $\tau = \frac{VQ}{Ib} = \frac{V}{2I} \left(\frac{h^2}{4} - y^2 \right)$)



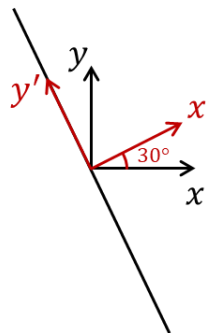
3. When water inside a cast-iron pipe freezes in winter, the pipe cracks due to the expansion of ice, but the ice itself is not damaged. This is because ().

- A. The strength of ice is higher than that of cast iron
- B. The ice is in a triaxial compressive stress state
- C. The temperature of the ice is higher than that of cast iron
- D. The stress in the ice is zero.

4. A structural element is subjected to a uniform plane-stress state denoted by σ_x , σ_y , τ_{xy} in the x - y coordinates. Assume $\sigma_x = -7$ MPa, $\sigma_y = 1$ MPa, and $\tau_{xy} = -3$ MPa.



- (1) Please determine the normal and shear stresses acting on the plane whose outward normal, x' -axis, is oriented at an angle to x -axis (see the figure below).



- (2) Please find the planes that are free of normal stress and determine the shear stress on these planes.
- (3) Please determine the principal stresses. Show them on a sketch of a properly oriented element.
- (4) Please determine the maximum shear stress and the associated normal stress. Show them on a sketch of a properly oriented element.

5. During a test of an airplane wing, the strain gage readings from a 45° rosette (see figure) are as follows:

$$\text{gage A, } 520 \times 10^{-6}$$

$$\text{gage B, } 360 \times 10^{-6}$$

$$\text{gage C, } -80 \times 10^{-6}$$

Determine the principal strains and maximum shear strains and show them on sketches of properly oriented elements.

(Hint: For a 45° rosette, $\varepsilon_x = \varepsilon_A$, $\varepsilon_y = \varepsilon_C$, $\gamma_{xy} = 2\varepsilon_B - \varepsilon_A - \varepsilon_C$)

